

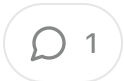
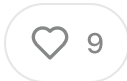
Why Your Volatility Forecast Matters More Than You Think

Introducing NextVoL®, our new framework for better volatility targeting



CONCRETUM RESEARCH

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Charlie Munger once summarized long-term investing in a single line:

“*The first rule of compounding is to never interrupt compounding unnecessarily.*”

In theory, this sounds trivial. In practice, compounding is rarely interrupted by a lack of return potential. It is interrupted by investors reacting to volatility. Share price drawdowns, violent daily swings, headline-driven regimes: these are the moments in which long-term investors quietly turn into short-term risk managers, risking at the worst possible time and abandoning plans they had every intention of following.

The lived experience of the market, the day-to-day path of outcomes, share price behavior. And once behavior becomes pro-cyclical, compounding becomes fragile.

This is precisely the problem volatility targeting is designed to address.

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Could you build a volatility forecasting methodology capable of estimating next-volatility with enough accuracy and stability to serve as the engine of an equity volatility-target index?

The motivation behind the question was not academic.

The underlying goal was to offer clients a better long-term investing experience: a smoother ride through turbulent markets, the kind of risk profile that helps investors stay committed to their plan rather than capitulate at the worst possible moment.

As is typical in the US market, they wanted to package the strategy into a capital-protected note where the end investor gains exposure to equity upside through a call option, typically with a one-year maturity. And in that context, the accuracy and stability of the volatility delivery directly determine the fair price of the option, which in turn determines the **participation rate** offered to the investor.

A more stable underlying means cheaper optionality, which means better terms for the end client.

That requirement, not merely forecasting volatility well but delivering a stable risk profile through a wide range of market regimes, is what drove the design of what would become the **NextVoL[®]** framework, developed jointly by Concretum|Research and Professors Walter Distaso and Antonio Mele.

The rest of this piece explains why standard forecasting methods struggle v

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The idea behind volatility targeting

The mechanics are simple. Rather than trying to forecast returns (famously difficult), a volatility-target strategy dynamically scales exposure to keep portfolio risk anchored around a predefined level.

The allocation rule is:

$$w_t = \frac{\sigma^*}{\hat{\sigma}_t}$$

where the numerator is the volatility target, while the denominator represents estimate of next-day volatility.

When expected volatility is below target, exposure is increased. When it is above target, exposure is reduced. If the volatility target is 10% and your forecast is 20%, you are 50% invested. That is it.

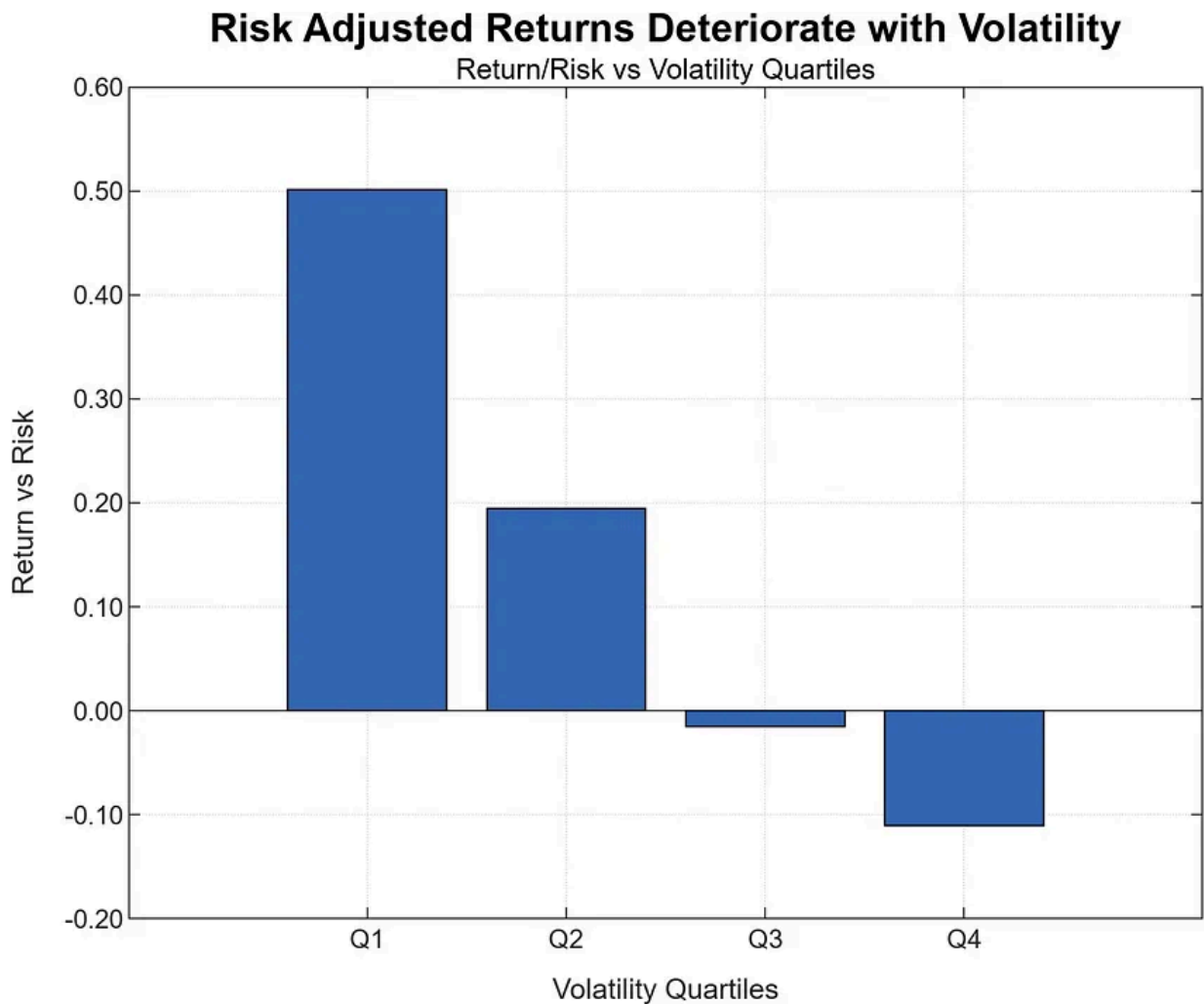
The appeal of this approach rests on two well-documented stylized facts.

First, **volatility is persistent**: high-volatility days tend to be followed by high volatility days, and calm clusters with calm.

Second, and more importantly for equity investors, **risk-adjusted returns tend to deteriorate as volatility increases**.

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Scaling exposure inversely with volatility is therefore not just a risk-stabilization overlay. It can improve long-run compounding efficiency.

But none of this works if the volatility forecast is poor. The scaling rule is mechanical. Its effectiveness rests entirely on the quality of the volatility estimate.

Why the forecasting method matters

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- **GARCH-type models.** Elegant econometrically, but sensitive to misspecification.
- **RiskMetrics EWMA.** Simple, but backward-looking with a fixed decay.
- **HAR models.** Multi-scale and parsimonious, but can lag at regime transitions.
- **VIX / implied volatility.** Forward-looking, but structurally biased upward by the volatility risk premium.
- **High-frequency realized volatility.** Responsive, but exposed to microstructural noise and unstable overnight adjustments.

Each of these approaches has strengths, but each also has a structural limitation that becomes visible when embedded in a volatility-target index.

Some models react too slowly and fail to stabilize risk during sharp regime shifts.

Others react too aggressively, generating excessive turnover and leverage instability.

Many rely on a *constant* scaling factor to approximate the overnight component of daily volatility, an assumption that does not hold when the relative contribution of overnight and intraday variance shifts across regimes.

The NextVol[®] framework

NextVol[®] is our attempt to address these structural limitations jointly, rather than one at a time.

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- **Short-horizon smoothing**, which attenuates the impact of transient volatility spikes and prevents the forecast from generating noisy day-to-day leverage swings.
- **Dynamic overnight adjustment**, which abandons the constant scaling multiplier and instead accounts for shifts in the relative weight of intraday versus overnight variance across regimes, reducing the systematic bias embedded in static corrections.
- **A stabilization mechanism**, which allows the forecast to adapt when realized risk persistently deviates from the objective, reinforcing consistent target delivery without sacrificing responsiveness.

The design philosophy is straightforward: keep the informational advantage of high-frequency data, but structurally discipline the well-known failure modes that come with it.

Empirical results

We tested all seven methodologies (the six standard approaches plus NextVoL[®] a 10% volatility-target index constructed on SPY, daily rebalanced, from January 2008 to December 2025).

No risk-free remuneration on the uninvested portion, no leverage cap. We deliberately stripped out frictions in order to isolate the pure effect of forecasting choice.

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Table 2: Performance summary of 10% volatility-target SPY indexes (2008–2025). TotRet, CAGR, Vol, MDD, AvgVol, and volatility percentiles are reported in percentage terms. Vol is the full-sample annualized realized volatility of the index computed from daily log returns. AvgVol is the time-series average of the rolling 252-day realized volatility. VVol denotes the standard deviation of the rolling 252-day volatility series. V_p denotes the p-th percentile of the rolling 252-day realized volatility series. MSE Rank ranks methods by MSE of rolling 252-day volatility relative to the 10% target (lower is better).

Method	TotRet	CAGR	Vol	Sharpe	MDD	MSE Rank	AvgVol	VVol	V_1	V_5	V_50	V_95	V_99
NextVoL [®]	320	8.31	10.16	0.79	-19.14	1	10.11	0.26	9.58	9.69	10.10	10.54	10.75
RV	302	8.04	10.21	0.76	-20.31	2	10.16	0.48	9.21	9.49	10.09	11.01	11.25
GARCH	284	7.76	9.88	0.76	-21.08	3	9.76	0.70	7.64	8.69	9.84	10.87	11.25
RiskMetric	301	8.03	10.58	0.73	-20.08	4	10.48	0.58	9.21	9.57	10.48	11.52	11.25
RV HAR	268	7.52	9.41	0.77	-18.87	5	9.35	0.58	8.35	8.53	9.26	10.40	10.75
HistVol	354	8.77	11.62	0.73	-21.05	6	11.50	0.68	10.05	10.33	11.43	12.71	12.75
VIX	179	5.88	7.94	0.72	-20.49	7	7.82	0.91	5.90	6.15	7.71	9.37	10.75
SPY	562	11.08	20.09	0.52	-51.85	–	17.83	8.32	6.86	9.80	15.38	34.67	45.75

A few observations stand out:

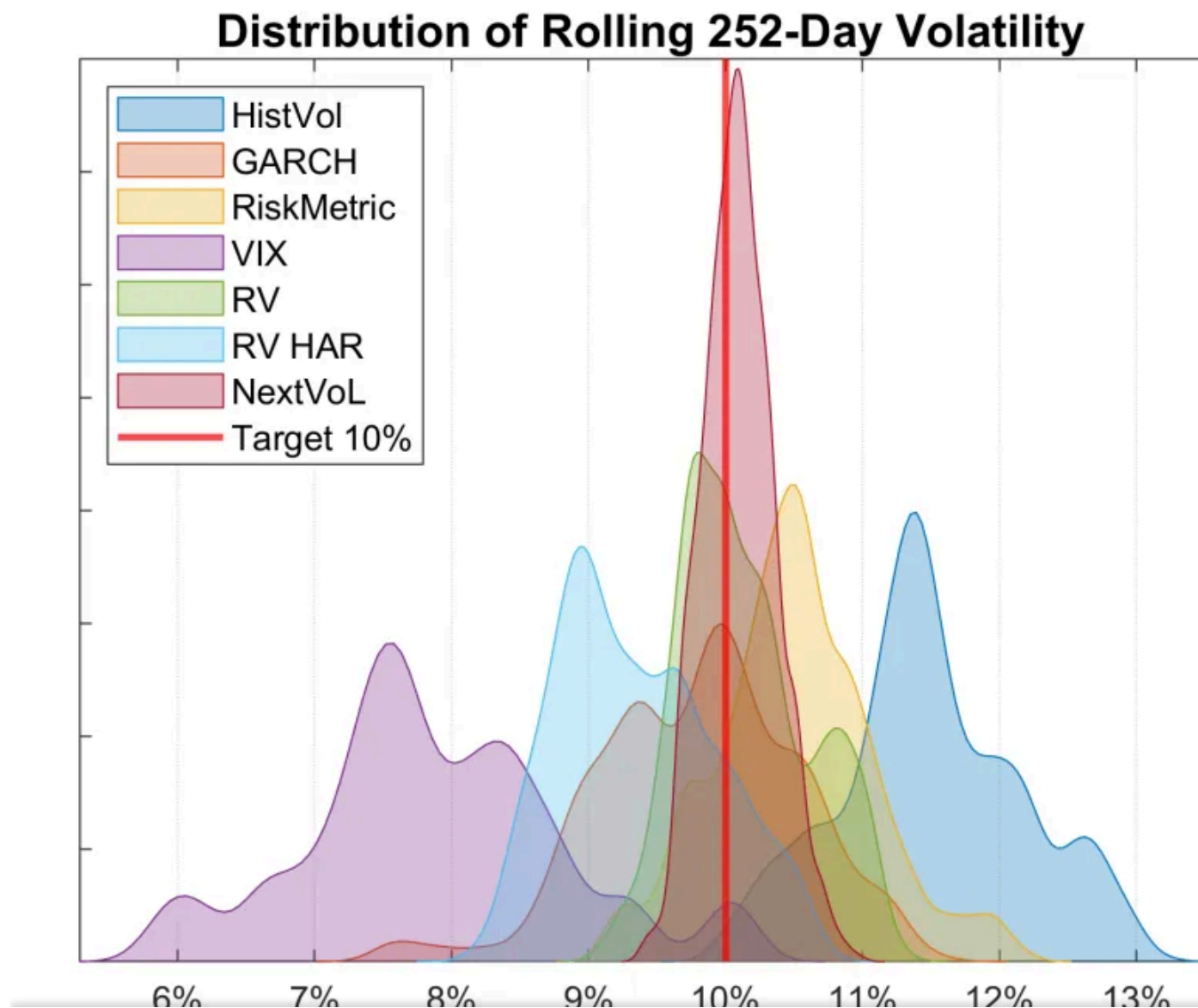
- **NextVoL[®] delivers the most accurate target**, with average rolling 252-day volatility of 10.11%, essentially on target, and the narrowest distribution of methods (1st percentile at 9.58%, 99th percentile at 10.75%).
- **Volatility-of-volatility is nearly half that of the next best method** (0.26 vs 0.48 for plain RV, 0.58 for HAR and RiskMetrics, 0.91 for the VIX-based approach).
- **The VIX-based strategy systematically undershoots the budget** (average realized vol of 7.82% against a 10% target), an under-utilization of roughly 23% driven by the embedded volatility risk premium.
- NextVoL[®] also produces the **highest Sharpe ratio (0.79)** among the volatility target approaches.

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wildly around it year after year, and a method that stays tightly anchored to target through time is a fundamentally different product from one that averages 10% by alternating phases of over- and under-delivery.

The full distribution of rolling realized volatility for each method makes this point visually:



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- **NextVoL®** is a tall, narrow spike centered almost exactly on the 10% target. It is the only distribution that does not meaningfully cross into either over- or under-delivery regions.
- The **VIX-based approach** sits far to the left, with its mode around 7.8%, a structural under-utilization of the volatility budget driven by the implied-volatility risk premium. Over an 18-year horizon, that is about 22% of the intended risk left on the table.
- **Historical Volatility** sits far to the right, with a wide distribution centered above 11.5%, systematically overshooting the target, and doing so inconsistently.
- The other methods (RV, RV HAR, GARCH, RiskMetrics) cluster reasonably well around the target but with distributions 2–3× wider than NextVoL®.

To rank the methods rigorously, we need a single summary statistic that penalizes **both** dimensions of miss at once: how far a method sits from the target on average, and how much it fluctuates around it through time. The natural candidate is the **mean squared error** of the rolling 252-day realized volatility relative to the 10% target:

$$\text{MSE} = \frac{1}{T} \sum_t (\sigma_t^{\text{VT}} - 10\%)^2$$

- The **squared deviation** penalizes bias. A method that averages 8% or 11.5% is structurally off-target, and that offset gets weighted in.
- It **also penalizes dispersion**. A method that averages exactly 10% but swings between 7% and 13% is not truly delivering the target either.

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Persistent, not sample-specific

The aggregate MSE ranking could, in principle, be driven by a handful favorable years.

To check for this, we decomposed the ranking year by year:

Table 3: Year-by-year MSE rank of rolling 252-day realized volatility relative to the 10% target (2008–2025). Each cell reports rank among the seven forecasting methodologies for a given calendar year (1 = best). AvgRank denotes the average of an ranks across all years.

Method	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	AvgRank
NextVoL®	2	1	1	1	1	1	1	1	3	1	2	1	1	1	1	1	1	3	1.3
RV	3	2	3	3	5	2	2	2	5	3	1	2	3	2	2	6	2	4	2.9
GARCH	6	5	2	2	2	4	4	4	2	5	4	3	2	3	3	3	4	2	3.3
RiskMetric	5	3	4	4	4	3	3	3	4	2	5	5	5	4	5	2	3	5	3.3
RV HAR	4	4	6	5	3	5	5	5	1	4	3	4	4	5	4	5	6	1	4.0
HistVol	7	7	5	6	6	6	6	6	6	6	6	7	7	6	6	4	5	7	6.0
VIX	1	6	7	7	7	7	7	7	7	7	7	6	6	7	7	7	7	6	6.0

NextVoL® ranks first or second in sixteen out of eighteen calendar years, with average annual rank of 1.3.

The methodologies that look close in the aggregate table (RV at 2.9 and GARCH at 3.3) turn out to be considerably more variable on a year-by-year basis. Both occasionally drop to rank 4, 5, or 6. This suggests that their aggregate performance partly reflects offsetting periods of over- and under-delivery, rather than consistent tracking. The NextVoL® advantage, by contrast, is structural rather than sample-specific, and that is precisely what you want from a methodology embedded in a live product.

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One could read the results above and focus on the Sharpe improvement. But Sharpe story, while real, is the less interesting half of the picture.

The more important advantage of NextVoL[®] is the **stability of volatility delivery itself**, captured by the vol-of-vol of 0.26 and the remarkably tight distribution of rolling realized vol around the 10% target. This is not a cosmetic property.

For anyone building **structured products, volatility-controlled indices, insurance-linked solutions such as Fixed Index Annuities (FIAs)**, the stability of the underlying's realized volatility is directly priced into the derivative overlay.

A lower vol-of-vol means lower uncertainty on future variance, which translates into **tighter option pricing on the wrapper**, cheaper hedges for the issuer, and crucially for the end investor, **higher participation rates** on the embedded optionality.

A volatility-target index that delivers its objective with a vol-of-vol of 0.26 provides a materially better foundation on which to build a structured product than one that delivers with 0.70 or 0.90. On paper, their average volatilities may look similar. In practice, the gap translates into sharper option pricing, higher participation rates, and a better deal for the investor holding the note.

If you are interested in understanding this study in greater detail, including the precise construction of the volatility-target indexes, the full empirical methodology, and the extended results across all forecasting approaches, we invite you to

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If you are working on volatility-controlled index design, structured product pricing, or volatility forecasting more broadly, we would love to hear your perspective. You can reach out at info@concretumgroup.com.

As always, feel free to leave a comment and reach out via direct message or email at info@concretumgroup.com for any questions.



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